COLLEGE OF ENGINEERING
BRADLEY DEPARTMENT OF ELECTRICAL AND COMPUTER ENGINEERING
VIRGINIA TECH

PacketVigil: AI Cyber Attack Detection

Sully Mrkva · Aiden Wang · Al Taber · Siri Rao · Joshua Manuel · Dheeraj Indela

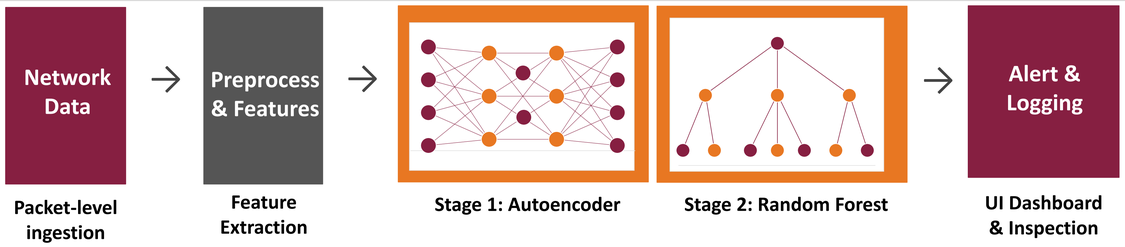
THE PROBLEM

Cyber threats are growing in complexity and frequency. **Signature-based IDS miss stealthy attacks** — they rely on known patterns, failing against novel attack vectors using non-standard ports or random intervals. Most open-source models and intrusion "detectors" operate exclusively offline. Models operate purely on offline datasets — not live data. Therefore, open-sourced real-time IDS systems are not only scarce—but critically needed.

OUR SOLUTION

A real-time, hybrid machine-learning-based detector combining an Autoencoder (anomaly detection) with a Random Forest classifier (attack classification), operating on live packet-level network traffic.

FULL SYSTEM PIPELINE



NETWORK INGESTION & FEATURE EXTRACTION ENGINE

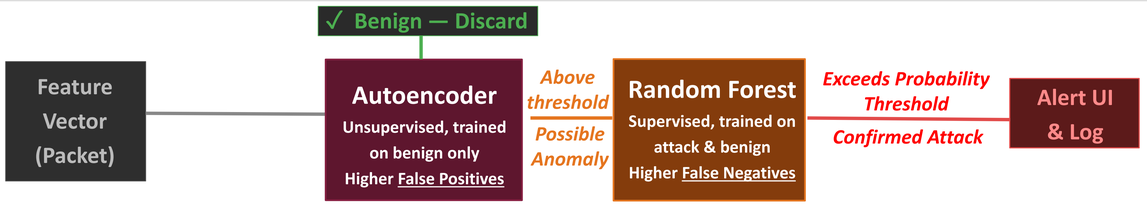
1 Packet Capture
Dumpcap
- High-performance capture engine

2 Packet Dissection
Dpkt
- Layer-by-layer parse
- Via binary pcap streams
- Ethernet - IP/ARP - TCP/UDP

3 Feature Extraction
Inspired by Kitsune (Mirsky et al. 2018)
- 115 behavioral features/pkt
- 5 traffic dimensions
- Exp. decay over 5 time-windows
- Cython-optimized

4 Flow Statistics & Inference
Behavioral vectors
- Packets not stored nor inspected
- Hash-tables updated/pruned
- Vectors sent for inference

ML CORE



Autoencoder *learns how to compress & reconstruct benign data, higher reconstruction error indicates anomaly.*

Random Forest *creates ensemble of decision trees to map features, classifies via weighted voting.*

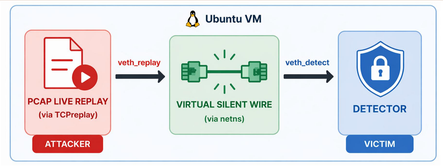
PERFORMANCE RESULTS – ON 6/7 DATASETS:

>90%	>95%	~1200	0.7ms
Recall	Precision	pkt/s Throughput	FE+ML latency/pkt

OFFLINE BENCHMARKING TESTBED



ONLINE TESTING SETUP

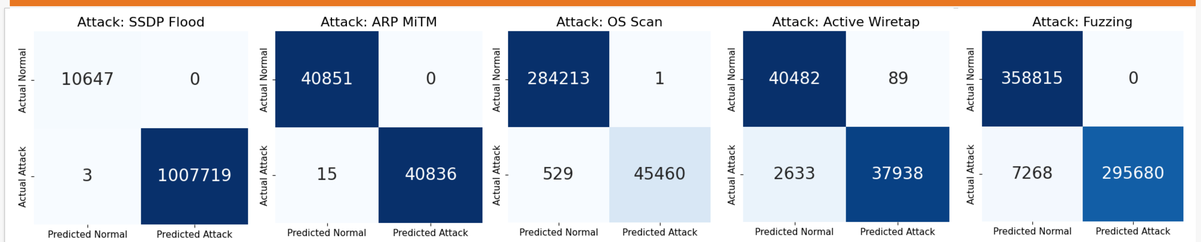


TESTING SUITE

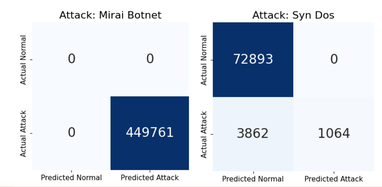
Attack Type	Man-in-the-Middle	Denial of Service	Reconnaissance	Malware
Description	Intercepts & manipulates network traffic	Overloads services to disrupt availability	Gathers information about hosts and services	Delivers malicious payloads
Datasets	Kitsune: **ARP MitM** Kitsune: **Active Wiretap**	Kitsune: **SSDP Flood** Kitsune: **SYN DoS**	Kitsune: **OS Scan** Kitsune: **Fuzzing**	Kitsune: **Mirai Botnet**

Kitsune datasets are open-source **PCAP** network captures with labeled attacks, generated from **real** host–attacker **interactions** using diverse tools.

CONFUSION MATRICES



AE reduces RF load as high-recall filter	
SSDP Flood	*Filters out **99%** benign flows*
OS Scan	*Filters out **85%** of benign flow*



Online Replay Snapshot: OS Scan	
Resource Usage	*On a 3 core VM, max **~570 MB** of Ram & **8%** of CPU*
Avg Throughput & Latency	***697** pps & **0.7**ms*

FUTURE WORK

*Our system is capable of **extracting** real-time network data and making very **accurate** detections on a per attack basis.*

- **Perform** robust cross-attack testing to evaluate full IDS generalization.
- **Explore** alternative decision logic approaches to fine-tune performance for cross-attack testing.
- **Implement** adaptive, real-time re-training at set intervals using our existing network data to training data scripts.
- **Optimize** AE structure, i.e., ensemble of autoencoders
- **Strengthen** feature extraction, less sensitive to timing/jitter artifacts
- **Research** protocol-specific feature extraction for richer traffic characterization.

Special thanks to our Customer – **Dr. Christiana Garcia**, Mentor **– Dr. Joe Adams, and Yisroel Mirsky et al --** for their open-sourced work on "*Kitsune…" NDSS '18*